

3-Colourability of Arrangements of Great Circles

OPG-34915

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Consider a set S of great circles on a sphere with no three circles meeting at a point. The arrangement graph of S has a vertex for each intersection point, and an edge for each arc directly connecting two intersection points. So this arrangement graph is 4-regular and planar.

Conjecture Every arrangement graph of a set of great circles is 3-colourable.